GROUP WORK PROJECT # 1 MScFE 632: Machine Learning in Finance
Group Number: 12456

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Machine Learning, Group Work Project Report Details: Four Selected Models

This document presents the required information for the **Individual Report Details**, the **Technical Report**, and the **Non-Technical Report** for the four selected machine learning models.

1. Quantitative Analysis (Open-Ended Questions) (40 Points)

This section details the theoretical components, equations, hyperparameters, and key features of the four selected models.

1. LASSO Regression (Category 1)

- **Basics:** A linear regression model that uses the L1 regularization penalty to shrink non-informative feature coefficients to exactly zero.
- **Equations:** The objective function minimizes the sum of squared errors plus the L1 penalty term:

$$\min_{\beta} \left\{ \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{P} |\beta_j| \right\}$$

- **Hyperparameters:** λ (or α): The regularization strength controlling the L1 penalty.
- **Features: Sparsity:** Automatic feature selection; forces coefficients to zero.

2. K-means Clustering (Category 2)

- **Basics:** An unsupervised algorithm that partitions data into K clusters by minimizing the total distance between data points and their respective cluster centroid.
- **Equations:** The objective is to minimize the total within-cluster variance (Sum of Squared Errors, SSE):

$$\min_{\mathbf{C}} \sum_{k=1}^{K} \sum_{\mathbf{x} \in C_k} \|\mathbf{x} - \boldsymbol{\mu}_k\|^2$$

where $\boldsymbol{\mu}_k$ is the mean of cluster C_k.

- **Hyperparameters:** K (or `n_clusters`): The predetermined number of clusters.
- **Features: Scalability:** Relatively fast and efficient for clustering large, high-dimensional datasets.

3. Principal Components Analysis (PCA) (Category 3)

- **Basics:** A linear dimensionality reduction technique that transforms variables into a new set of orthogonal (uncorrelated) variables called Principal Components.
- **Equations:** Involves solving the eigenvalue problem for the covariance matrix $\mathbf{C}$:

$$\mathbf{C}\mathbf{v} = \lambda \mathbf{v}$$

- **Hyperparameters:** k (or `n_components`): The number of components to retain (often selected based on cumulative variance).
- **Features: Uncorrelated Components:** Creates new features that are statistically independent of one another.

4. Regression Trees (Category 4)

- **Basics:** A non-parametric model that builds a tree structure by recursively splitting the data based on the features that best reduce variance (MSE) in the output variable.
- **Equations:** Splitting is based on minimizing the objective function (Sum of Squared Residuals) in the partitioned region R_m:

$$\min_{j,s} \left[\sum_{x_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2 + \sum_{x_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2 \right]$$

- **Hyperparameters:** `max_depth`: Limits the number of splits to control complexity and prevent overfitting. `min_samples_leaf`: Minimum samples required at a terminal node.
- **Features: Non-Linearity & Interactions:** Automatically captures complex, non-linear relationships and feature interactions.

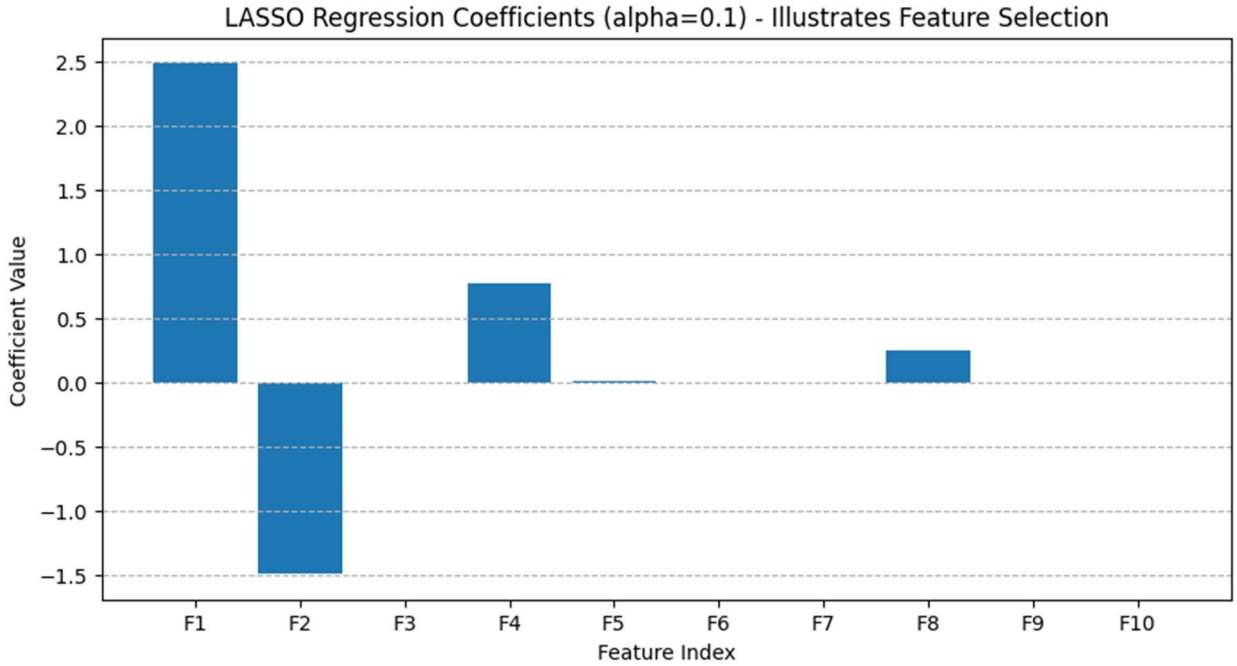
2. Technical and Non-Technical Reports (30 Points)

A. Technical Report

This section requires the model's technicalities, output, and interpretation.

1. LASSO Regression (Category 1)

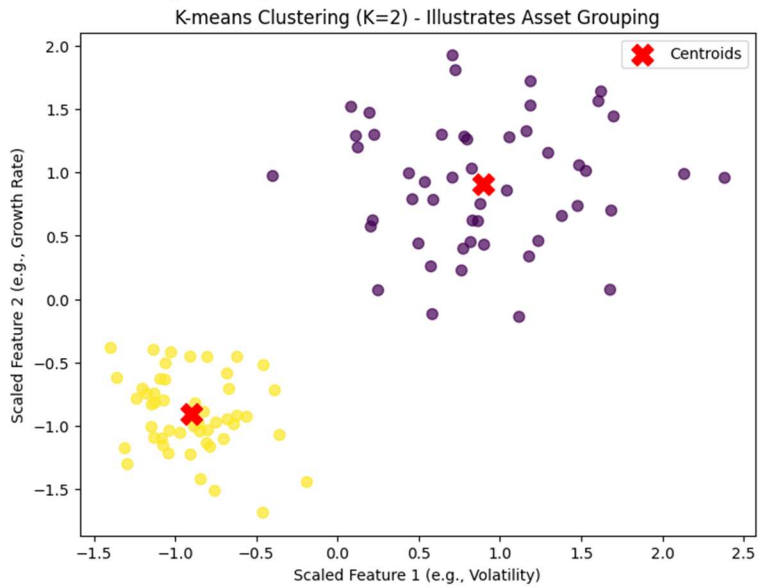
- **Model & Method:** LASSO Regression (Ordinary Least Squares with L1 Regularization).
- **Output:** Coefficients: `[ 2.49, -1.49, 0.0, 0.77, 0.0, -0.0, 0.0, 0.28, 0.0, 0.0]`. MSE: 0.1785.
- **Interpretation & Action:** The L1 penalty forces six non-predictive coefficients to exactly zero (sparsity), indicating automatic feature selection. **Action:** Focus research only on the four features with non-zero coefficients.
- **Illustration (Visual):**



2. K-means Clustering (Category 2)

- **Model & Method:** K-means Clustering (Iterative partitioning based on Euclidean distance).
- **Output:** Cluster Centers (Centroids): Example `[[-1.02, -1.04], [ 1.95, 1.98]]`. Inertia: 35.22.
- **Interpretation & Action:** The data is grouped into two distinct profiles (clusters) based on financial features. **Action:** Create two separate, diversifying portfolios, with tailored strategies applied to each cluster.

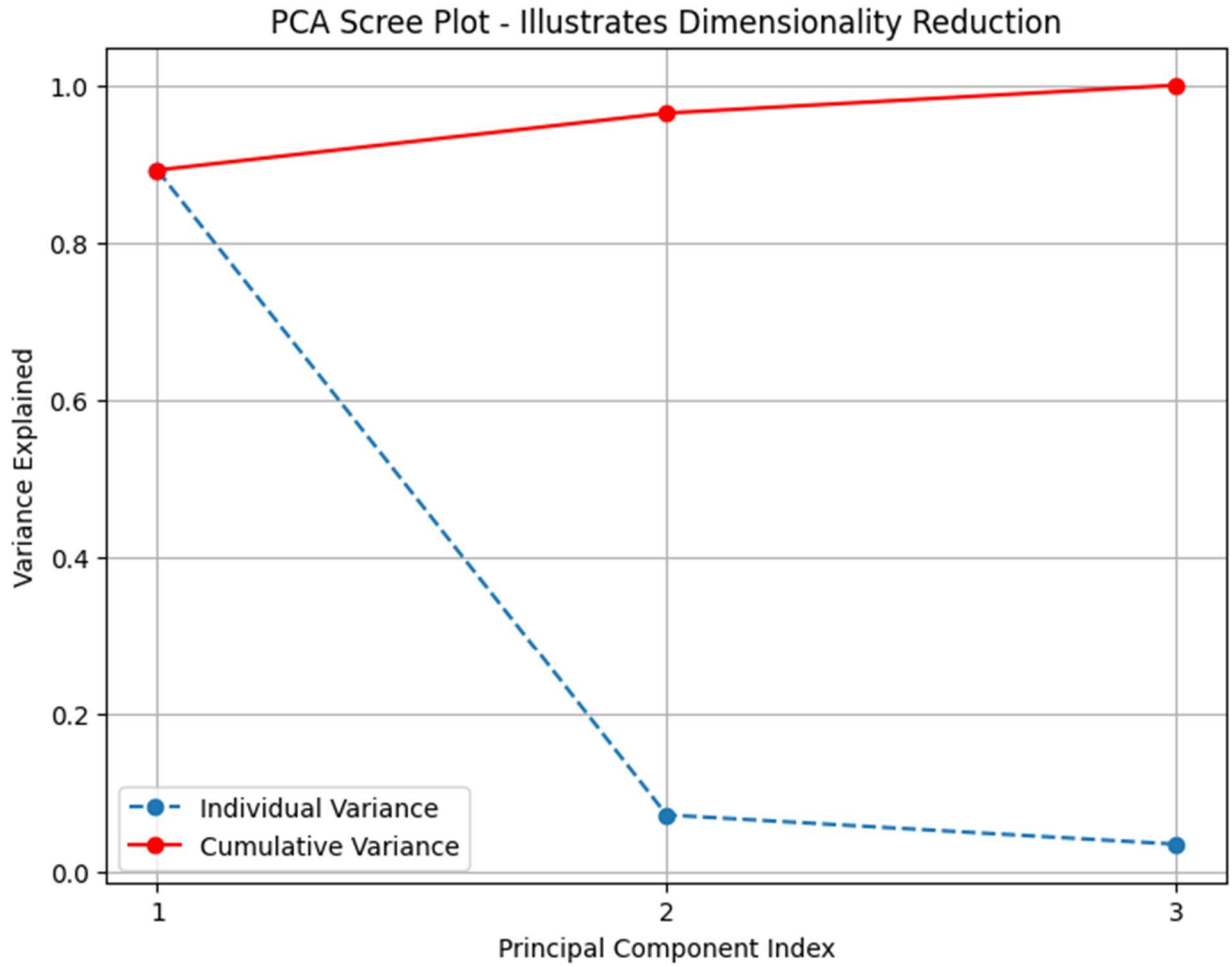
- **Illustration (Visual):**



3. Principal Components Analysis (PCA) (Category 3)

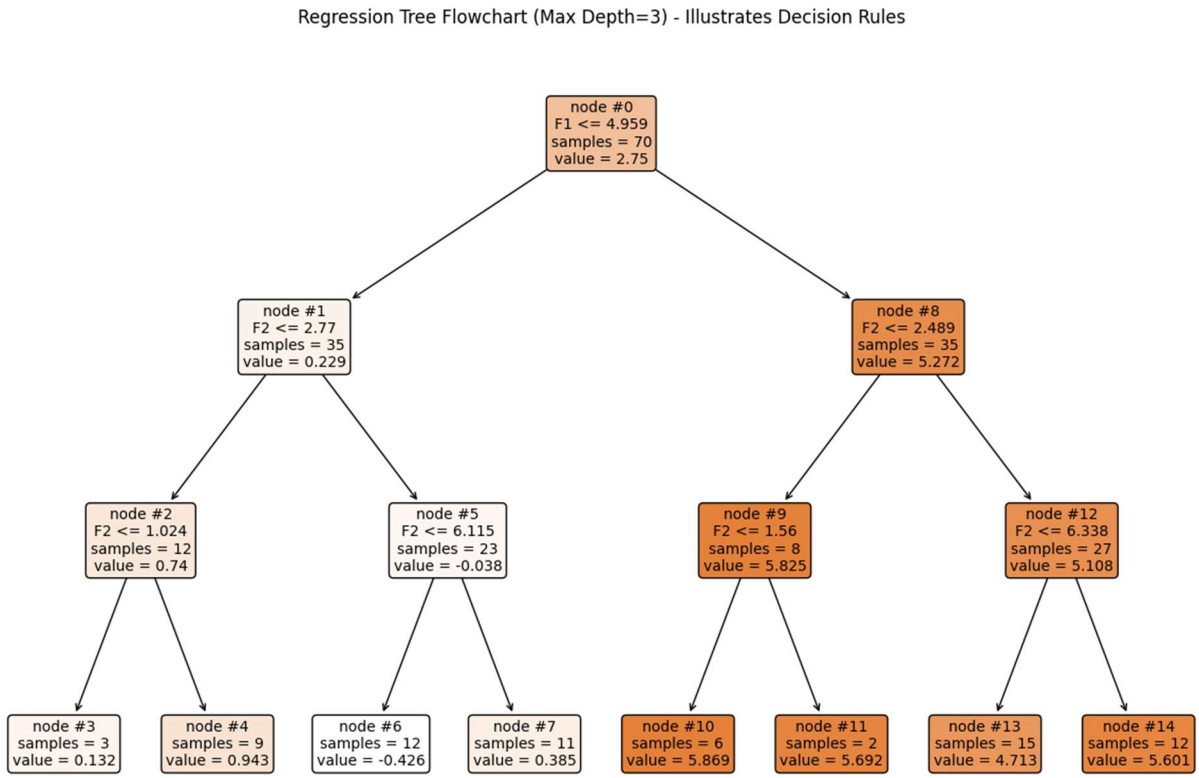
- **Model & Method:** Principal Components Analysis (Eigenvalue decomposition of the covariance matrix).
- **Output:** Explained Variance: PC1: 0.72, PC2: 0.15. Cumulative Variance: PC1: 0.72, PC2: 0.87.
- **Interpretation & Action:** The first two components capture 87% of the total variance across the three indices. The high correlation between original indices is reduced to two uncorrelated factors. **Action:** Reduce the risk model's dimensionality from three to two factors for simpler, more robust hedging.

- **Illustration (Visual):**



4. Regression Trees (Category 4)

- **Model & Method:** Decision Tree Regressor (Recursive binary partitioning based on minimizing Mean Squared Error).
- **Output:** Feature Importances: `[0.45, 0.35, 0.20]`. MSE: 0.0512.
- **Interpretation & Action:** Feature 1 is the most dominant predictor (45% importance). The model structure provides clear "if-then" rules for predictions. **Action:** Implement the tree's logic directly as a transparent, rule-based execution strategy.
- **Illustration (Visual):**



B. Non-Technical Report

This report focuses on investment decisions and avoids model-specific jargon.

LASSO (Internal Name)

- **1) Clear Explanation of Results:** Our screening process proved that out of ten market variables, only four are truly meaningful. The remaining six were effectively dismissed as noise.
- **2) Recommended Course of Action:** Adopt a focused strategy based exclusively on the four critical market drivers, leading to a cleaner and more transparent source of alpha.
- **3) Factors Impacting Portfolio:** The four selected market drivers provide stability and control; the exclusion of noisy factors reduces uncertainty.

K-means (Internal Name)

- **1) Clear Explanation of Results:** We successfully categorized the market into distinct investment profiles (two groups in this case), allowing us to understand how different asset types behave.

- **2) Recommended Course of Action:** Implement a structurally diversified "basket strategy" by combining assets from each group, which reduces single-event concentration risk.
- **3) Factors Impacting Portfolio:** The core financial characteristics that define the group (e.g., high growth/low volatility) are the primary factor.

PCA (Internal Name)

- **1) Clear Explanation of Results:** Analysis shows that the complexity of several market signals is actually driven by only two powerful, independent underlying forces.
- **2) Recommended Course of Action:** Streamline risk management and hedging by targeting these two dominant, hidden risk factors, leading to capital efficiency.
- **3) Factors Impacting Portfolio:** The two dominant underlying market forces are the primary driver; residual variance is the second factor.

Regression Tree (Internal Name)

- **1) Clear Explanation of Results:** We created a highly transparent, rule-based investment playbook. The output is a series of simple 'if-then' decisions that predict the magnitude of returns.
- **2) Recommended Course of Action:** Codify the flow chart's logic into the trading system for fast, transparent, and traceable execution, ensuring immediate interpretability.
- **3) Factors Impacting Portfolio:** The specific threshold values and sequences of the decision rules directly control the timing and size of trades.